

Program: Diploma in Fire Technology and Safety	
Course Code: 5462	Course Title: Safety in Oil & Gas Industries
Semester: 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand the different types of hazards in the oil and gas industry, including chemical, toxic, explosion, and mechanical hazards.
- To learn about the safety aspects related to petroleum hydrocarbons, sour gases, and their effects on human health.
- To explore safety management practices, environmental considerations, and offshore safety protocols.
- To familiarize students with various risk analysis techniques and emergency management plans for both onshore and offshore operations.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand various hazards in chemical and petroleum industries and their prevention.	12	Understanding

CO2	Analyze the effects of petroleum hydrocarbons, sour gases, and related accidents.	15	Analyzing
CO3	Learn safe storage and handling practices, safety audits, and environmental safety management.	14	Applying
CO4	Understand risk analysis methods, emergency plans, and safety systems in the oil & gas sector.	17	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3					
CO3	3	3	2				
CO4	3	3	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand various hazards in chemical and petroleum industries and their prevention.		
M1.01	Setting and layout of a chemical plant for safety.	4	Understanding
M1.02	Forms of hazards in the industry: chemical, toxic, explosion, electrical, mechanical, radiation, and noise hazards.	4	Understanding
M1.03	Control and prevention of hazards in chemical and petroleum plants.	4	Applying

Contents: Safety in the Chemical and Petroleum Industry

Safe layout and design considerations for chemical and petroleum plants.

Identifying different hazards in the chemical and petroleum industry.

Control and prevention strategies for various industrial hazards (e.g., chemical, electrical, noise)..

CO2	Analyze the effects of petroleum hydrocarbons, sour gases, and related accidents.		
M2.01	Toxicity, asphyxiation, and respiratory effects of petroleum hydrocarbons and sour gases.	4	Understanding
M2.02	Skin effects and threshold limits of exposure.	4	Understanding
M2.03	Analysis of documented accidents in the petroleum industry, including drilling, fracturing, and blowouts.	4	Analyzing
M2.04	Corrosive atmosphere in the oil field and its effects on operations.	3	Understanding
	Series Test I	1	

Contents: Toxicity, Asphyxiation, and Respiratory Safety in Petroleum Operations

Health risks from petroleum hydrocarbons and sour gases.

Effects of exposure to petroleum hydrocarbons, including skin and respiratory risks.

Review and analysis of real-life accidents in petroleum engineering.

CO3	Learn safe storage and handling practices, safety audits, and environmental safety management.		
M3.01	Characteristics of chemicals with reference to safe storage and handling.	2	Understanding
M3.02	Layout of storage facilities for petroleum products and associated hazards.	2	Applying
M3.03	Modes of transport for hazardous materials and associated hazards.	2	Understanding
M3.04	Safety audit procedures, engineering standards, and regulatory agencies.	3	Applying
M3.05	Offshore safety: procedures for managing oil spills and oil spill control.	3	Applying

M3.06	Safety and Environmental Management Systems (SEMS).	2	Understanding
Contents: Safe Storage, Handling, and Environmental Management Safe storage, handling, and transportation of petroleum products. Safety audits, regulatory standards, and the role of agencies in managing petroleum operations. Offshore safety and oil spill control systems. Introduction to Environmental Management Systems (SEMS) in the oil and gas industry			
CO4	Understand risk analysis methods, emergency plans, and safety systems in the oil & gas sector.		
M 4.01	Risk analysis techniques: HAZOP studies, hazard analysis (HAZAN), fault tree analysis, and consequence analysis	3	Applying
M 4.02	Scenario and probabilistic assessments for risk evaluation	2	Analyzing
M 4.03	Onshore and offshore emergency management plans.	2	Understanding
M4.04	Safety systems: manual and automatic shutdown systems, blow-down systems, and gas detection systems.	3	Applying
M4.05	Fire detection and suppression systems in petroleum operations.	3	Understanding
M4.06	Personal protective systems and measures in the oil and gas industry.	2	Understanding
M4.07	HSE Policies and disaster/crisis management in the petroleum industry.	2	Analyzing
	Series Test II	1	
Contents: Risk Analysis and Emergency Management in Oil & Gas Understanding different risk analysis methods (HAZOP, HAZAN, fault tree analysis). Designing and implementing emergency management plans. The role of safety systems such as shutdown systems, blow-down systems, and gas detection systems. Fire detection and suppression measures. Personal protective systems and their use in protecting workers. Health, Safety, and Environmental (HSE) policies and crisis management in the petroleum industry.			

Text /Reference:

T/R	Book Title/Author
R1	Crowl, D.A., Louvar, J.F., Chemical Process Safety – Fundamentals with Applications, Prentice-Hall, Pearson, 2011.
R2	CCPS, Guidelines for Engineering Design for Process Safety, AIChE.
R3	Kletz, T.A., Hazop and Hazard Analysis, IChemE, 2001.
R4	Mannan, M.S., Lee's Loss Prevention in the Process Industries, Elsevier, 2012.
R5	Murray, M., & Green, D., Offshore Safety Management Systems, Gulf Publishing, 2008.
R6	Lund, L. et al., Environmental Protection in the Oil and Gas Industry, Springer, 2014.